

Customer Satisfaction and Sentiment Analysis



Data Understanding

	id_survey	date_of_survey	ticket_system	overall_rating	customer_service	features	value_for_money	ease_of_use	likelihood_to_recommend	overall_text
1	T_00229	2024-10-06	Zendesk	3.00	4.00	3.00	3.00	2.00	6.00	Customer tickets managements
2	T_04527	2024-12-26	Zoho Desk	5.00	5.00	5.00	5.00	5.00	8.00	After 6 months of using the Zoho desk, we shif...
4	T_00644	2024-10-17	Zendesk	5.00	3.00	4.00	5.00	5.00	6.00	Pros:Zendesk has always been one of the go-to ...
6	T_04682	2024-12-28	Zoho Desk	5.00	4.00	5.00	5.00	5.00	8.00	It has been very useful so far to integrate mu...
8	T_01238	2024-11-02	Freshdesk	4.00	4.00	4.00	5.00	4.00	8.00	Pros:It's easy to use and very intuitive.We ha...



The dataset is a customer satisfaction survey consisting of 787 complete records.



Customer ratings include customer service, features, value for money, and ease of use, which are averaged into an overall rating (scale 1–5).



The dataset also includes likelihood to recommend, representing customers' willingness to recommend the product (scale 1–10).



An overall review text is provided and can be used for sentiment analysis.

Data Preprocessing



Missing Value Check

Out of 1,482 records, only 787 were complete, so the rest were removed.



Duplicated Value Check

No duplicated value was found.



Converting Data Types

date_of_survey data type is converted from object to datetime.



Out-of-scale Value Check

20 values were out of scale (e.g., -1), so they were removed.

Customer Satisfaction Analysis

Three customer satisfaction analysis methods are used:



Customer Satisfaction Analysis

Result:

	csat	customer_service_satisfaction	features_satisfaction	value_for_money_satisfaction	ces	nps_category
1	91.18	67.34	88.34	87.60	89.48	Detractor
2	91.18	67.34	88.34	87.60	89.48	Passive
4	91.18	67.34	88.34	87.60	89.48	Detractor
6	91.18	67.34	88.34	87.60	89.48	Passive
8	91.18	67.34	88.34	87.60	89.48	Passive



NPS categories were derived from segmentation of likelihood_to_recommend scores

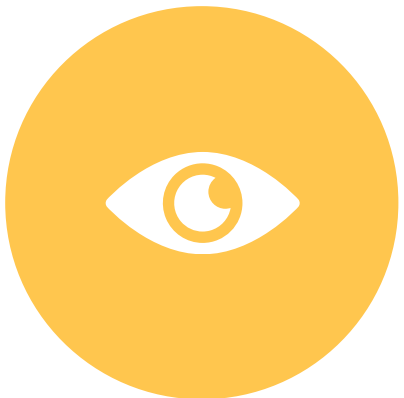


The resulting NPS score is 11.94%, which falls into the Average category

Customer Sentiment Analysis



A RoBERTa model is used for customer sentiment analysis.



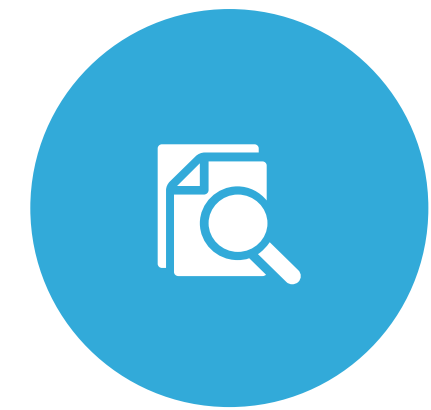
RoBERTa (Robustly Optimized BERT Pretraining Approach) is an improved BERT-based language model trained on large-scale data to better understand text context, making it suitable for customer sentiment classification.

The Output Includes:



Sentiment score

0–1



Sentiment labels

Positive, Neutral,
Negative

Customer Sentiment Analysis

Result:

	sentiment	sentiment_score	sentiment_label
1	LABEL_1	0.73	Neutral
2	LABEL_2	0.98	Positive
4	LABEL_2	0.79	Positive
6	LABEL_2	0.96	Positive
8	LABEL_2	0.87	Positive

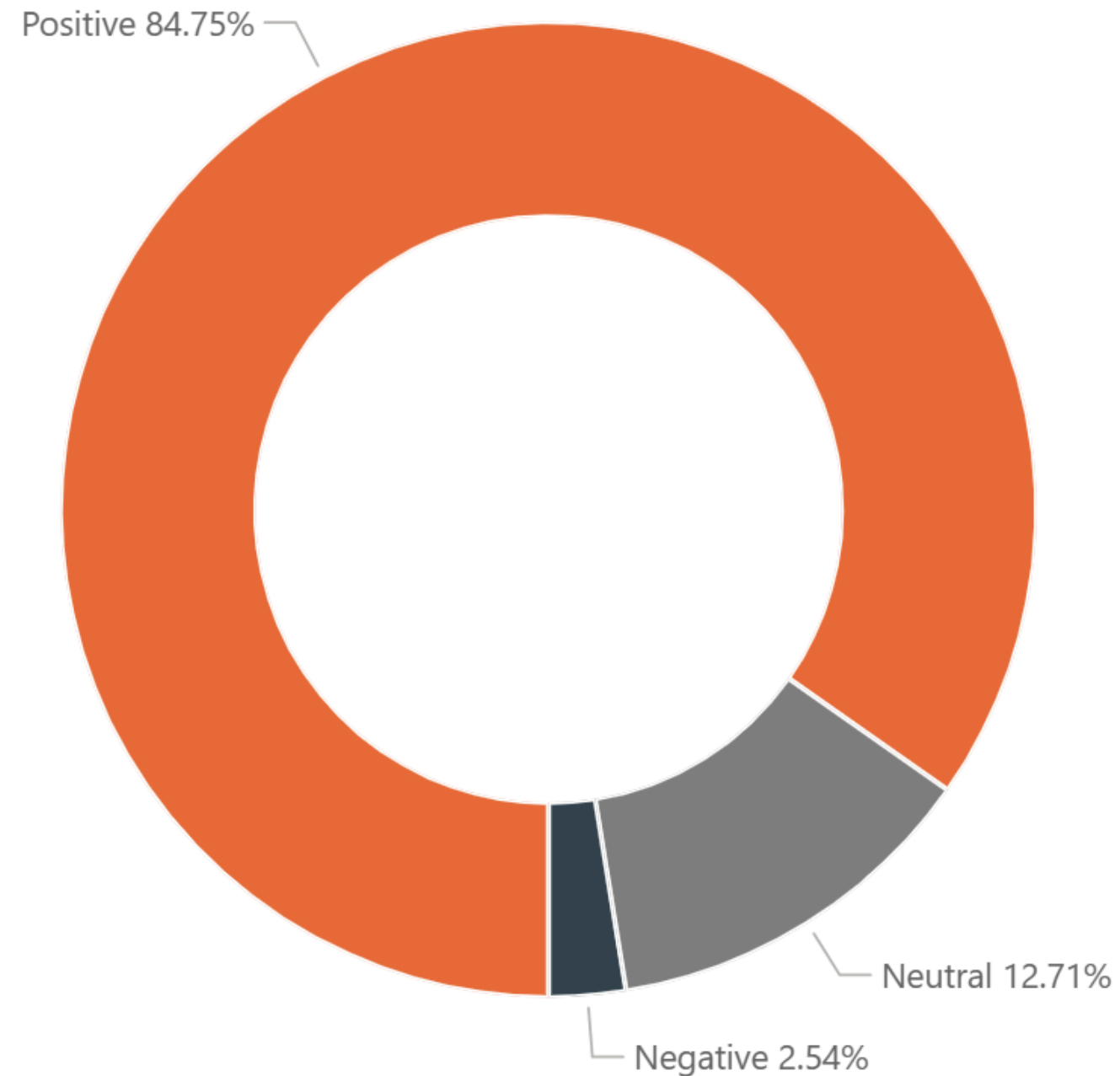


RoBERTa models use default labels Label 0, Label 1, and Label 2.



These labels are then mapped to Negative, Neutral, and Positive.

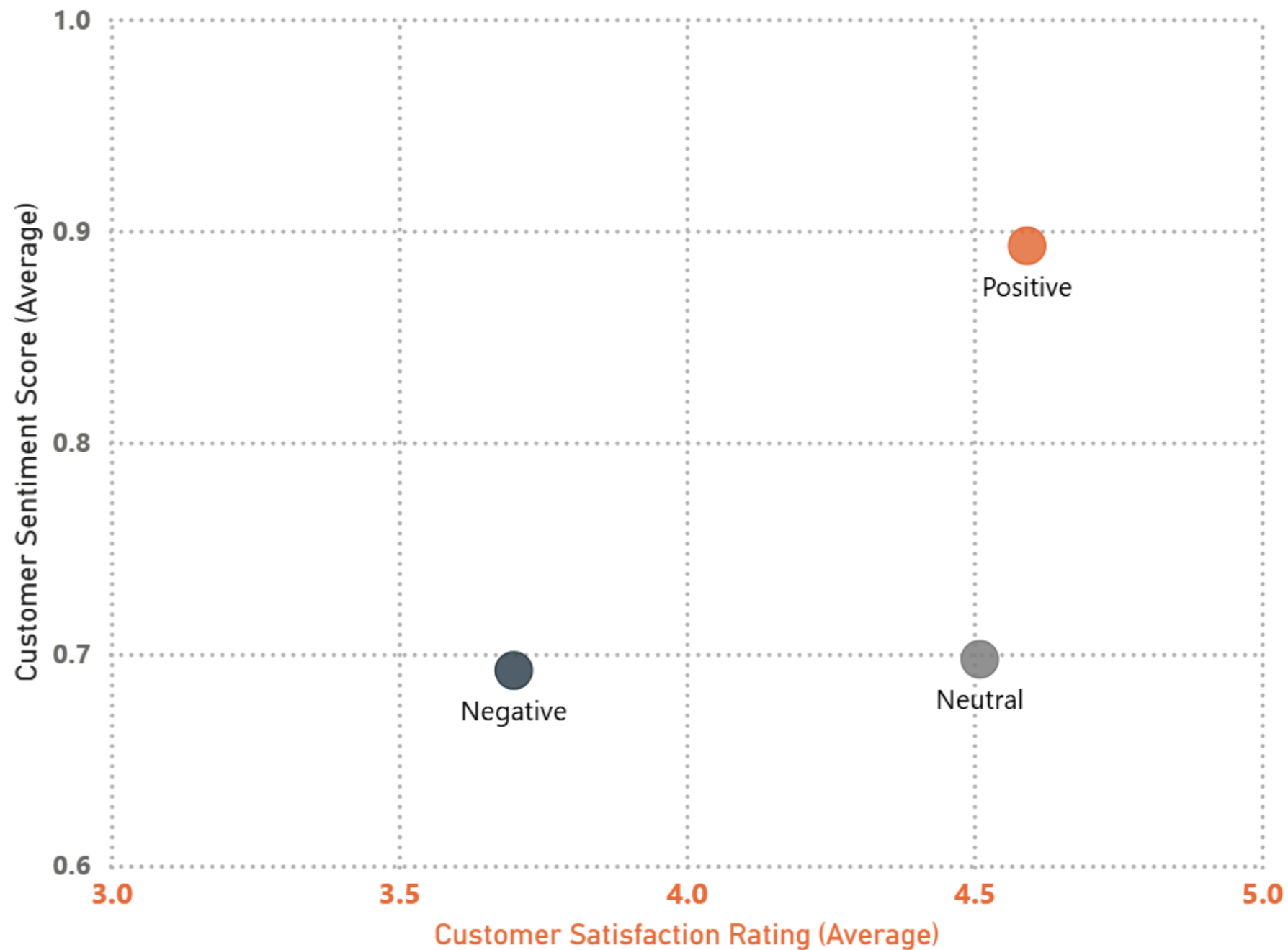
Customer Sentiment Distribution



Out of 787 surveyed customers, 84.75% (667) provided positive reviews, while approximately 15% (100 Neutrals + 20 Negatives) still gave non-positive feedback.

This indicates that there is still room for improvement in customer service, features, value for money, and ease of use.

Correlation Between Customer Sentiment Scores and Ratings



The scatter plot shows that customers giving low ratings (below 4) tend to have negative sentiment scores.



Customers with ratings between 4 and 5 generally show neutral sentiment scores.



Customers giving ratings close to 5 have positive sentiment scores.



Conclusion: Customer satisfaction ratings are positively correlated with their sentiment scores.

Recommendations



Since customer satisfaction ratings and sentiment scores are positively correlated, improving the scores of the five key metrics (Customer Service, Features, Value for Money, Ease of Use, and Overall Experience) will likely increase both satisfaction and sentiment scores.

Enhance Customer Service

Provide faster responses, improve support quality, and train staff to handle complaints effectively.

Improve Product Features

Identify commonly requested features or pain points and prioritize their implementation.

Increase Value for Money

Offer promotions, bundles, or improve product quality to better match customer expectations.

Simplify Ease of Use

Optimize user interface and experience to make the product/service more intuitive.

Sentiment Tracking

Regularly track sentiment scores alongside ratings to quickly identify areas needing improvement.